Supplementary Material: Executable Evidence Management for CAD-Guided Aircraft Inspection

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S1. Extended numerical methods and evidence properties

This section accompanies the revised main article. It retains the detailed numerical interface and the conditions supporting the execution and evidence properties. The subsequent protocol and result catalogues retain their original separate study identities.

S1.1. Joint-state propagation and checkpoint covariance

The 18-dimensional local perturbation includes robot translation and rotation, six motion biases, and map translation and rotation. Propagation uses

$$x_{k+1} = f(x_k, u_k), \qquad P_{k+1} = F_k P_k F_k^\top + Q_k. \tag{S1}$$

The bias follows a first-order Gauss–Markov model. Robust registration uses a fixed source prior, Huber residuals, an unmatched-point penalty and backtracking fractions $1, 1/2, 1/4, 1/8, 1/16$. The checkpoint covariance is

$$P_s^{(j)} = \mathrm{PSD}\left[(G_j^\top (P_s^-)^{-1} G_j + H_j^\top \Omega_j H_j)^{-1}\right]. \tag{S2}$$

The Jacobians include local rotation perturbations. This covariance is a curvature approximation; empirical risk calibration uses complete executions. A final rejection after an eligible interim publication emits an ordered original-prior rollback. Source replacement begins replay from that source candidate's state and covariance, using only acquired increments. The one-active-source service drains computation and pending publication before admitting another source.

S1.2. Installation work and selection

For publication j, let c_j be total sender work including the common front end, $r_{<j}$ accumulated receiver work, r_0 fixed receiver work and h propagation work per tick. With queue delay $q = 0.05$ s, time step $\Delta t = 0.1$ s and load α, the controlled installation clock uses

$$n_j = \max\left(1, \left\lceil \frac{q + \alpha(c_j + r_{<j} + r_0)}{\Delta t - \alpha h} \right\rceil\right), \qquad r_{<j+1} = r_{<j} + r_0 + n_j h. \tag{S3}$$

The denominator is positive for the frozen services. Next-stage reservation uses the common upper operation count before starting a registration iteration. The sender cap includes numerical and decision work; common front-end, stream and receiver work remain in total availability and work. Mean-characterized operation coefficients define this simulation service. Component process CPU and whole-online-route CPU are reported separately.

The typed record is $\mathcal{K} = (\mathcal{V}, \tau, \mathcal{R}, \sigma, \Gamma)$: entities, entity types, relations, immutable specifications and executable predicates. A before-sensing request $R = (g, a, \mathcal{E}, \delta)$ fixes the population, service, event set and allowance. Selection uses

$$\Pi(R) = \{\pi \in \Pi_0 : \mathrm{match}(g, a, \pi, \nu) \wedge U_{g,a,\pi,e} \le \delta \text{ for all } e \in \mathcal{E}\}, \tag{S4}$$

$$\pi^*(R) = \arg\min_{\pi \in \Pi(R)} \left(\overline{L}_{g,a,\pi} + \lambda_{\mathrm{sel}} \overline{W}_{g,a,\pi}\right). \tag{S5}$$

Policy-ID order resolves ties. An empty set refuses statistical admission. Precommit, actual source sealing, version checking, replay and claim issuance execute on the selected path. The original numerical service remains available with provisional evidence status after detected invalid attachment history.

S1.3. Retained fitted models and policy controls

The retained CART/RF fit uses 926 training and 194 validation sources from a 1,120-source development set. Causal inputs are prior uncertainty, weakest scaled information eigenvalue, residual/inlier measures, step sizes, objective ratio, iteration, pair count, displacement and declared noise. Reachable terminal caps are $\{2, 4, 8, 16, 25\}$. Heuristic predictions estimate terminal MSE, development-event frequency, remaining work and current-versus-prior MSE. Effective terminal labels include rejection and rollback. Publication compares eligible candidate MSE with the prior; selecting an already reached cap from iteration two stops computation. Empty time-feasible sets or profile constraints may also stop the job.

The 22-policy catalogue contains contract-CART, FlatCART, FlatRF and ContractRF at prices 0.3/1/3; Fixed2/4/8/16, Final, Residual, Random2or4 and OldRAVC; and price-one CART/RF time-only interventions. KG denotes the archived contract-CART name. Time-only controls add the same deadline predicate to the flat forecast; the contract adds the declared-profile predicate next. All use the shared stage budget wrapper and acquisition/replay semantics. The time screen predicts delivery using conditional mean remaining work and candidate replay, omitting previously accumulated receiver work; actual event scoring uses the full installation clock. The profile screen caps refinement at four iterations when the declared noise multiplier exceeds two, source position standard deviation exceeds 0.061 m, or geometry was excluded from predictor fitting.

OldRAVC uses the earlier six-output depth-six CART, fitted on 480 training and 120 validation sources. It retains the binary C172 feature. An eligible candidate publishes when forecast event frequency is at most 0.30 and forecast MSE improves on the prior. From iteration two, it stops when predicted four-second MSE gain is no greater than remaining-work cost at price 1 m^2, with the same event-frequency screen. Residual stops at the first eligible checkpoint from iteration two. Random2or4 chooses cap four with frozen continuation probability 0.4654427646, otherwise cap two, using the archived seed-hash coin. Its iid binomial interpretation includes an ideal independent randomization stream. Fixed caps publish an eligible reached cap; Final waits for solver termination. Budget interruption is included in their effective execution maps.

S1.4. Source replacement and delivery exposure

Proposition 1 (Source replacement)**.** *Assume an immutable scan/prior, a retained solve unchanged by publication and no interleaved independent update. Installing the final candidate at a fixed time gives the same live state and covariance whether intermediate candidates were delivered or the final candidate was delivered alone. The same statement holds for ordered prior rollback.*

Proof. The final candidate is the same function of the scan, prior and executed prefix in both schedules. Both receivers assign its replay using the same acquired motion. Intermediate versions contribute no second likelihood factor. □

Proposition 2 (Delivery exposure)**.** *For schedules sharing candidates, exogenous motion and final installation τ_f, let one publish an intermediate candidate at $\tau_c < \tau_f$ and the other hold it. Under integrability,*

$$\mathbb{E}[L_{\text{hold}} - L_{\text{publish}}] = \int_0^T \mathbb{E}\big[\mathbf{1}\{\tau_c \le t < \tau_f\}\left(\|e_0(t)\|^2 - \|e_c(t)\|^2\right)\big]\, dt. \tag{S6}$$

Proof. Estimates coincide before intermediate installation and after the aligned final installation. Subtract the loss integrals pathwise and exchange integration and expectation. □

Eligibility alone does not determine the sign of the exposure difference. Replay has cost. The executed held-publication intervention recomputes delivery timing and may change final installation, so its empirical contrast includes that change.

S1.5. Simultaneous coverage and attachment noninterference

Theorem 3 (Finite-family population-event coverage)**.** *Freeze K complete policy/population/service/event combinations before their calibration outcomes. Assume iid complete inputs from the deployment mixture for each event, including independent policy randomization, and fixed effective execution maps. Exact one-sided binomial bounds with undercoverage allocation β/K cover every risk simultaneously with probability at least $1-\beta$. On that confidence event, calibration-dependent selection among the admitted combinations satisfies its requested risk allowances.*

Proof. Each fixed combination has a binomial failure count and exact inversion undercoverage at most β/K. A union bound establishes simultaneous coverage. Every admitted selection then has its required bounds on that event. Dependence across methods, services and events on a shared input is permitted. □

Confidence probability concerns uncertainty over calibration; the allowance concerns a deployment event. A directly calibrated union differs from two marginal checks. For M episodes the generic mission bound is only $\min(1, M\delta)$ absent further structure. Geometry subgroups and receipt-conditioned subsets require their own statements.

Proposition 4 (Attachment noninterference)**.** *Let $\mathcal{P}$ project a canonical execution $B(X)$ onto numerical decisions, produced/installed versions, replay/rollback order, modeled work, loss, readiness and events. Suppose an observer W calls the original validation/installation with the same arguments and results, changes only separate claim state and excludes its native timing from the frozen modeled clock. Then*

$$\mathcal{P}(W(B(X))) = \mathcal{P}(B(X)). \tag{S7}$$

Canonical valid histories retain original receipts. A detected invalid attempt permanently removes the upper bound in that source scope; a later valid numerical version may still install.

Proof. Induction over validation and installation preserves every original numerical/clock step. Separate claim updates change none of the projected state. The history latch is a persistent conjunction, so a failed attempt cannot reset it within the source scope. □

This projection preserves the canonical calibrated event map. It does not calibrate an arbitrary faulty-history population or a subset selected by remaining receipts. A changed modeled installation schedule requires new calibration.

S1.6. Requirement closure and producer transfer

Proposition 5 (Coverage of a requirement closure)**.** *On the simultaneous confidence event of fixed anchors, an anchor A whose success set is contained in that of R gives $\Pr(E_{p,R}) \le U_{p,A}$. This holds for every requirement in the anchor's pointwise containment closure.*

Proof. Success-set containment gives failure-set containment $E_{p,R} \subseteq E_{p,A}$. Taking probabilities and applying the simultaneous anchor bound proves the statement. Derived requirements add no union-bound allocation. □

Proposition 6 (Invariant requested projection)**.** *Suppose a valid old anchor covers R, the complete-input population is unchanged and every requested function satisfies $z_{p',j}(X) = z_{p,j}(X)$ for all supported inputs. Then the old anchor also bounds $\Pr(E_{p',R})$ on the same confidence event.*

Proof. Requested-function equality yields identical request events, including nonfinite-output failure semantics. Combine that equality with old-producer containment and the anchor bound. □

The premise is functional invariance in the trusted implemented model. Independent A/B pools are used only for such invariant events and retain both source-block identities. Before-sensing commitment is essential: choosing after observing the source between two disjoint 0.10-risk events can produce a selected event with risk 0.20. A directly calibrated switching policy or a common covering event would be needed for that alternative.

S1.7. Unique event support and analytical report timing

The revised main article adds two descriptive checks without changing frozen labels, cards or decisions. Deduplication uses (study, population, service, event_id). Each 192-label catalogue contains 160 unique event contexts. Contract A/B support is 24/23; direct-renewal support is 30/29; exact-card support is 18/17. Label support remains 24/26, 30/32 and 18/20, respectively.

Report timing satisfies $a_{\text{image}} \geq 0.05 + \alpha w_{\text{camera}}$ and $t_{\text{report}} \geq c + 0.05 + \alpha w_{\text{camera}}$. With frozen camera work of 0.0625 s in A and 0.125 s in B, the image-age lower bounds are A: 0.675/1.3/2.55/3.8 s and B: 1.3/2.55/5.05/7.55 s across relaxed/standard/tight/contended services. Five combinations necessarily violate the full-report 1.8-s image-age limit. Remaining combinations still require acquisition, association and actual timing success.

S2. Study identities, sampling and prospective records

The numerical localization study and the two image-task studies use distinct independently generated inputs and calibration families. A source seed is the sampling unit. Repeated policies, services, events and requests are paired executions of that source. The change/recovery validation is a deterministic software study on existing evidence; it contributes no new population samples or calibration cards.

Table S1: Independent source blocks. Nominal and unfitted populations have disjoint seed ranges. Each stress block contains four explicitly unsupported strata.

Study	Calibration	Confirmation	Stress	Reserved event family
Localization	8,192	4,096	1,536	528
Image A, 192×144	4,096	2,048	256	4,608 jointly
Image B, 288×216	4,096	2,048	256	

The localization source blocks start at 8,300,000/8,400,000 for calibration and 8,500,000/8,600,000 for confirmation. Its four 384-source stress strata start at 8,700,000, 8,800,000, 8,900,000 and 9,000,000. The 48 excluded pilot sources start at 8,200,000. The retained predictor fit has 926 training and 194 validation sources.

For image A, calibration starts at 13,000,000/13,100,000 and confirmation at 13,200,000/13,300,000. Image B uses 14,000,000/14,100,000 and 14,200,000/14,300,000, respectively. Each image population contains 2,048 calibration and 1,024 confirmation sources. Stress starts at 13,400,000 or 14,400,000, with an additional 1,000 per successive stratum and 64 inputs per stratum. The 48 excluded image pilots start at 12,000,000. There are 409,600 paired policy/service outcomes and 6,291,456 repeated confirmation requests across A/B; these counts do not increase the independent source count.

Nominal draws mix B767/C172, position-prior deviations 0.015/0.03/0.06 m, range-noise multipliers 1/2 and shape levels 0/1/2. Source frames are sampled from prescribed trajectories, excluding the first and last 100 frames. Map translation and rotation perturbations have standard deviations 0.025 m and 0.001 rad. The unfitted procedural airframe is absent from predictor training and has independently matched calibration. Stress separately removes a 2.2-rad range sector, retains a 0.5-rad sector, increases prior deviation to 0.08/0.12 m, or introduces a 0.06-m map warp.

Numerical methods, generators, fitted predictors, event definitions and characterized clocks were sealed at F0 before formal generation. Calibration cards and before-sensing plans were sealed at F1 before confirmation. Versioned amendments are preserved: localization source-history monitoring changed only attachment state, with audited numerical/clock/event noninterference; image manifest formatting preserved scientific parameters; conservative attachment dependencies added capture/motion/geometry providers; direct-renewal parity restored access to applicable old evidence. The final image family remains 4,608 and unused allocation remains reserved. The archived hashes establish local execution chronology.

Two missing localization index records were recovered on their original seeds with identical numerical arrays; their recovered CPU fields are new timing measurements. Continuous routes contain 36 distinct 600-s

inputs, with four matched realizations for each geometry/condition and 12 methods, yielding 432 executions. The online timer excludes generation, archive hydration and truth scoring. The retained original manuscripts, superseded protocols and earlier A19/A20/R4/R5 results are separately archived; they do not enter the current sample counts or supplement tables.

S3. Image requirements, dependencies and timing

The acquisition event includes fixed-target GSD, exposure blur and acquired target coverage. Association includes coverage, reprojection, metric displacement and a positive version at capture. Reporting combines both tasks with delay and image age. Their limits define synthetic diagnostic services; palette landmarks evaluate acquisition and association within the implemented image producer.

Table S2: Canonical full, partial and strict thresholds. Blank bounds denote a quantity omitted from that service. Reprojection and metric association errors retain different units.

Quantity	Full	Partial	Strict
GSD (mm/pixel)	20	40	10
Blur (pixel)	0.50	0.75	0.35
Coverage (fraction)	0.50	0.20	0.70
Reprojection (pixel)	30	50	15
Association (m)	0.10	0.15	0.06
Report delay (s)	2	3	1
Image age (s)	1.8	2.8	0.8

Partial-relaxed removes GSD and reprojection bounds and uses delay/age 4/3.8 s. Coarser-GSD, more-blur and later-report variants change only the named full threshold to 40 mm/pixel, 0.75 pixel or 4/3.8 s. Acquisition's component variant retains GSD 40 and blur 0.75; association's retains displacement 0.15 m and readiness; reporting's retains delay/age 3/2.8 s. These eight named variants contain duplicates for tasks unaffected by a particular edit. There are 192 labels but 160 unique population/service/event contexts per study at each allowance.

The compiler supports length and time conversions, pixel units and ratios. It rejects incompatible dimensions, Boolean/nonfinite thresholds, duplicate quantity intervals and empty intervals. Nonfinite required outputs fail an event. An absent required output raises a schema error and prevents valid evaluation. Thus missing schema fields are distinguished from a defined observation's failure. Preflight checks 10,000 event pairs on one million generated metric vectors and 144 task-specific fixtures; those checks are excluded software cases.

Every requested quantity includes the conservative capture/motion/geometry-provider dependencies from the sealed attachment amendment. The other dependencies follow the executed data flow: GSD includes target plan and focal length; blur additionally includes exposure; coverage includes ROI, rendering, pixel noise and decoding; reprojection/association additionally include localization; readiness includes localization; report delay/age include localization, virtual camera clock and queue. Transfer checks the union of requested dependencies, retaining old calibration lineage. Changes in quantities outside that requested projection do not enter its capsule.

Both ROIs use focal length 1,800 pixels, exposure 6 ms and the same preplanned target definition. The excluded-pilot virtual camera/decoder clocks are 0.0625 s for A and 0.125 s for B. With load 10/20/40/60, image-age lower bounds are A: 0.675/1.3/2.55/3.8 s and B: 1.3/2.55/5.05/7.55 s. Five study/service combinations necessarily violate the full 1.8-s limit. Remaining combinations still require all other task conditions.

The image receipt observer operates on archived numerical records and is timed separately. Its source seal, study, population, policy, event, plan and card checks preserve source-scoped refusal across reconstructed gates. Original fixtures include 72 provider edits and 32 malformed packets. The original source replacement/replay checks execute inside the numerical localization service. Component CPU, whole-route CPU, virtual operation

work and preloaded-record query wall time retain separate boundaries. Windows camera CPU has 15.625-ms quantization.

S4. Inference, sensitivity and data interfaces

Exact binomial inversion uses β/K, with $\beta = 0.05$, $K = 528$ for localization and $K = 4,608$ jointly for A/B. A valid statistical statement assumes iid complete inputs, a fixed executed event map and matched deployment distribution. Correlation across policies, events or services sharing a source is permitted by the simultaneous union-bound argument. Pooling A/B is restricted to explicitly invariant requested projections and preserves both independent manifest identities.

The 16 primary localization contrasts use 2,048 confirmation sources per population and 32,000 paired bootstrap resamples. Pointwise and family-adjusted percentile intervals are approximate sampling intervals. They are distinct from exact binomial admission bounds. Nested prefixes, alternative confidence levels, geometry subgroups, Pareto fronts, stress and complete routes are descriptive; they do not revise F1 or obtain additional conditional guarantees.

The accompanying CSV files retain full-precision cards, all 6,144 plans, confirmation summaries, all geometry/service/policy means, unsupported stress, route results and software case decisions. Printed tables focus on the current research questions. Complete population source generation and raw arrays remain in the original verified release assets. The manuscript package includes sealed evidence, five complete archived replay inputs, table snapshots, deterministic figure scripts, the typed compiler, producer-audit and case-validation code, and SHA-256 manifests. Submitted figures use original photographs, executed synthetic rasters and deterministic vector graphics.

S5. Executed change and recovery validation

The supplementary software study uses all 320 unique primary A/B request contexts. The original semantic policy is held fixed; a direct-renewal policy supplies the fixed alternative when that semantic request was refused, otherwise Fixed2 is the prespecified control. Sixteen configurations per context comprise no change, each of 13 single provider changes, and ROI/clock and decoder/clock pairs. For each method, the validator checks the edited context and restoration of the original trusted producer. An independently written oracle inspects intervals and the same amended dependency specification without calling the hash or containment routines used by the implementations.

Three complete methods receive identical applicable card inventories and cached evidence: contract coverage, independently coded procedural coverage and cached direct renewal. Stale-plan reuse and whole-producer invalidation are separately identified diagnostic controls. Incorrect retention and unnecessary withdrawal are defined relative to the same method's admissible archived evidence. These are deterministic conformance counts, with zero new geometric draws and zero new calibration cards. A restored producer uses its original specification and a fresh trusted execution scope after a detected history fault.

The concrete source is the previously archived replay seed 14,300,000, selected in the earlier reproduction handoff. Its RGB array, numerical state arrays and execution metadata are hash checked. FlatCART at relaxed service installs iterations one and two at 0.1/0.2 s, and capture at 0.2 s uses the second version. The full-association anchor has $k/n = 229/2,048$ and $U = 0.1439731071$. A predeclared reprojection relaxation from 30 to 50 pixels lies in its containment closure. A decoder edit withdraws full association while preserving a covered readiness-only requirement. The optical component remains refused under the unfitted-population risk bound. Restoring the original decoder restores the full-association plan; packet history faults remain refused until a fresh trusted execution scope.

The first illustrative assertion expected the optical component to be admitted. Its actual archived bound rejected that expectation. The failed source/freeze and refinement note are retained, the optical refusal stays visible, and the readiness-only derived requirement was added to demonstrate existing component evidence under the unchanged requested projection. The 320-by-16 change matrix, numerical observations and risk allocations were unchanged. Query timing covers preloaded-card revalidation and restoration only.

Matched-support record counts measure the effort to construct the required proof catalogues; warmed caches in all complete methods require zero further record rescoring.

S5.1. Conformance and common-support accounting

Table S3: Executed deterministic software decisions over the same prespecified change contexts. Counts are conformance fixtures, not independent population samples or fault-rate estimates.

Method	Decisions	Missed	Unnecessary	Restore
Contract	5120	0	0	0
Cached direct	5120	0	0	0
Complete procedure	5120	0	0	0

Table S4: Matched support with the same precommitted policies. Counted work to construct selected proof catalogues from archived calibration records; it excludes source acquisition and producer execution. Both complete implementations share anchor accounting.

Study	Method	Requests	Cards	Record visits
A	Anchor reuse	24	12	24576
A	Direct events	24	15	30720
B	Anchor reuse	23	11	24576
B	Direct events	23	14	30720
combined	Anchor reuse	47	23	49152
combined	Direct events	47	29	61440

S6. Complete localization confirmation catalogue

Table S5: All 22 policies, both populations and all four services. Loss L is 10^{-3} m^{2}s; modeled work W is ms; failure columns are percentages. Every source is retained. Native CPU and full tails are in the accompanying CSV.

Population	Service	Policy	n	L	W	Obs.	Union
unfitted	contended	CART_Time	2048	22.353	7.616	77.25	77.98
unfitted	contended	ContractRF	2048	22.405	7.869	92.43	92.77
unfitted	contended	ContractRF0.3	2048	22.405	7.869	92.43	92.77
unfitted	contended	ContractRF3	2048	22.405	7.869	92.43	92.77
unfitted	contended	Final	2048	18.559	12.001	100.00	100.00
unfitted	contended	Fixed16	2048	18.559	12.001	100.00	100.00
unfitted	contended	Fixed2	2048	20.763	9.453	100.00	100.00
unfitted	contended	Fixed4	2048	18.559	12.001	100.00	100.00
unfitted	contended	Fixed8	2048	18.559	12.001	100.00	100.00
unfitted	contended	FlatCART	2048	18.173	11.199	79.15	79.69
unfitted	contended	FlatCART0.3	2048	17.967	11.653	79.15	79.64
unfitted	contended	FlatCART3	2048	19.078	10.148	79.15	79.64
unfitted	contended	FlatRF	2048	20.523	10.344	92.72	92.92
unfitted	contended	FlatRF0.3	2048	20.523	10.344	92.72	92.92
unfitted	contended	FlatRF3	2048	20.523	10.344	92.72	92.92

Population	Service	Policy	n	L	W	Obs.	Union
unfitted	contended	KG0.3	2048	22.353	7.616	77.25	77.98
unfitted	contended	KG1	2048	22.353	7.616	77.25	77.98
unfitted	contended	KG3	2048	22.353	7.616	77.25	77.98
unfitted	contended	OldRAVC	2048	18.611	10.139	82.91	83.06
unfitted	contended	RF_Time	2048	22.405	7.869	92.43	92.77
unfitted	contended	Random2or4	2048	19.659	10.623	100.00	100.00
unfitted	contended	Residual	2048	20.417	9.853	76.81	77.44
unfitted	relaxed	CART_Time	2048	17.095	10.598	8.01	8.79
unfitted	relaxed	ContractRF	2048	17.086	11.054	8.35	8.94
unfitted	relaxed	ContractRF0.3	2048	16.671	11.917	8.25	8.94
unfitted	relaxed	ContractRF3	2048	18.503	10.063	8.89	9.62
unfitted	relaxed	Final	2048	17.030	11.688	8.74	9.47
unfitted	relaxed	Fixed16	2048	17.030	11.688	8.74	9.47
unfitted	relaxed	Fixed2	2048	19.937	9.219	9.13	10.06
unfitted	relaxed	Fixed4	2048	17.030	11.688	8.74	9.47
unfitted	relaxed	Fixed8	2048	17.030	11.688	8.74	9.47
unfitted	relaxed	FlatCART	2048	17.095	10.598	8.01	8.79
unfitted	relaxed	FlatCART0.3	2048	16.859	10.998	8.20	8.89
unfitted	relaxed	FlatCART3	2048	18.129	9.667	8.54	9.23
unfitted	relaxed	FlatRF	2048	17.086	11.054	8.35	8.94
unfitted	relaxed	FlatRF0.3	2048	16.671	11.917	8.25	8.94
unfitted	relaxed	FlatRF3	2048	18.503	10.063	8.89	9.62
unfitted	relaxed	KG0.3	2048	16.859	10.998	8.20	8.89
unfitted	relaxed	KG1	2048	17.095	10.598	8.01	8.79
unfitted	relaxed	KG3	2048	18.129	9.667	8.54	9.23
unfitted	relaxed	OldRAVC	2048	17.717	9.718	8.35	9.03
unfitted	relaxed	RF_Time	2048	17.086	11.054	8.35	8.94
unfitted	relaxed	Random2or4	2048	18.535	10.348	9.13	10.06
unfitted	relaxed	Residual	2048	19.609	9.388	8.98	9.91
unfitted	standard	CART_Time	2048	17.423	10.702	7.91	8.79
unfitted	standard	ContractRF	2048	19.936	9.953	9.08	10.06
unfitted	standard	ContractRF0.3	2048	19.936	9.953	9.08	10.06
unfitted	standard	ContractRF3	2048	19.936	9.953	9.08	10.06
unfitted	standard	Final	2048	17.350	11.736	8.54	9.28
unfitted	standard	Fixed16	2048	17.350	11.736	8.54	9.28
unfitted	standard	Fixed2	2048	19.938	9.220	9.08	10.06
unfitted	standard	Fixed4	2048	17.350	11.736	8.54	9.28
unfitted	standard	Fixed8	2048	17.350	11.736	8.54	9.28
unfitted	standard	FlatCART	2048	17.423	10.702	7.91	8.79
unfitted	standard	FlatCART0.3	2048	17.194	11.113	8.15	8.89
unfitted	standard	FlatCART3	2048	18.431	9.750	8.54	9.23
unfitted	standard	FlatRF	2048	19.936	9.953	9.08	10.06
unfitted	standard	FlatRF0.3	2048	19.936	9.953	9.08	10.06
unfitted	standard	FlatRF3	2048	19.936	9.953	9.08	10.06
unfitted	standard	KG0.3	2048	17.194	11.113	8.15	8.89
unfitted	standard	KG1	2048	17.423	10.702	7.91	8.79
unfitted	standard	KG3	2048	18.431	9.750	8.54	9.23
unfitted	standard	OldRAVC	2048	17.964	9.781	8.40	9.03
unfitted	standard	RF_Time	2048	19.936	9.953	9.08	10.06
unfitted	standard	Random2or4	2048	18.675	10.375	9.03	9.96
unfitted	standard	Residual	2048	19.857	9.469	8.94	9.91
unfitted	tight	CART_Time	2048	22.172	7.552	94.87	95.12
unfitted	tight	ContractRF	2048	22.195	7.795	99.95	99.95
unfitted	tight	ContractRF0.3	2048	22.195	7.795	99.95	99.95
unfitted	tight	ContractRF3	2048	22.195	7.795	99.95	99.95

Population	Service	Policy	n	L	W	Obs.	Union
unfitted	tight	Final	2048	22.168	7.542	94.53	94.78
unfitted	tight	Fixed16	2048	22.168	7.542	94.53	94.78
unfitted	tight	Fixed2	2048	22.168	7.542	94.53	94.78
unfitted	tight	Fixed4	2048	22.168	7.542	94.53	94.78
unfitted	tight	Fixed8	2048	22.168	7.542	94.53	94.78
unfitted	tight	FlatCART	2048	22.172	7.552	94.87	95.12
unfitted	tight	FlatCART0.3	2048	22.172	7.552	94.87	95.12
unfitted	tight	FlatCART3	2048	22.172	7.552	94.87	95.12
unfitted	tight	FlatRF	2048	22.195	7.795	99.95	99.95
unfitted	tight	FlatRF0.3	2048	22.195	7.795	99.95	99.95
unfitted	tight	FlatRF3	2048	22.195	7.795	99.95	99.95
unfitted	tight	KG0.3	2048	22.172	7.552	94.87	95.12
unfitted	tight	KG1	2048	22.172	7.552	94.87	95.12
unfitted	tight	KG3	2048	22.172	7.552	94.87	95.12
unfitted	tight	OldRAVC	2048	22.171	7.549	94.78	95.02
unfitted	tight	RF_Time	2048	22.195	7.795	99.95	99.95
unfitted	tight	Random2or4	2048	22.168	7.542	94.53	94.78
unfitted	tight	Residual	2048	22.168	7.542	94.53	94.78
nominal	contended	CART_Time	2048	17.620	7.583	49.22	49.95
nominal	contended	ContractRF	2048	17.679	7.839	66.16	66.60
nominal	contended	ContractRF0.3	2048	17.679	7.839	66.16	66.60
nominal	contended	ContractRF3	2048	17.679	7.839	66.16	66.60
nominal	contended	Final	2048	12.823	12.377	100.00	100.00
nominal	contended	Fixed16	2048	12.823	12.377	100.00	100.00
nominal	contended	Fixed2	2048	14.802	9.432	100.00	100.00
nominal	contended	Fixed4	2048	12.823	12.377	100.00	100.00
nominal	contended	Fixed8	2048	12.823	12.377	100.00	100.00
nominal	contended	FlatCART	2048	12.527	11.183	50.88	52.39
nominal	contended	FlatCART0.3	2048	12.170	12.053	50.88	52.49
nominal	contended	FlatCART3	2048	13.098	10.098	50.88	52.05
nominal	contended	FlatRF	2048	14.577	10.351	66.85	67.82
nominal	contended	FlatRF0.3	2048	14.577	10.351	66.85	67.82
nominal	contended	FlatRF3	2048	14.577	10.351	66.85	67.82
nominal	contended	KG0.3	2048	17.620	7.583	49.22	49.95
nominal	contended	KG1	2048	17.620	7.583	49.22	49.95
nominal	contended	KG3	2048	17.620	7.583	49.22	49.95
nominal	contended	OldRAVC	2048	13.106	10.007	56.54	57.03
nominal	contended	RF_Time	2048	17.679	7.839	66.16	66.60
nominal	contended	Random2or4	2048	13.608	10.756	100.00	100.00
nominal	contended	Residual	2048	14.481	9.838	48.78	50.00
nominal	relaxed	CART_Time	2048	11.408	10.619	5.76	6.93
nominal	relaxed	ContractRF	2048	11.125	10.990	5.66	6.64
nominal	relaxed	ContractRF0.3	2048	10.706	12.446	6.05	7.13
nominal	relaxed	ContractRF3	2048	12.152	10.031	5.81	7.23
nominal	relaxed	Final	2048	10.911	12.072	6.79	7.81
nominal	relaxed	Fixed16	2048	10.911	12.072	6.79	7.81
nominal	relaxed	Fixed2	2048	13.830	9.221	7.08	8.59
nominal	relaxed	Fixed4	2048	10.911	12.072	6.79	7.81
nominal	relaxed	Fixed8	2048	10.911	12.072	6.79	7.81
nominal	relaxed	FlatCART	2048	11.408	10.619	5.76	6.93
nominal	relaxed	FlatCART0.3	2048	11.002	11.389	5.86	6.88
nominal	relaxed	FlatCART3	2048	12.031	9.646	5.81	7.08
nominal	relaxed	FlatRF	2048	11.125	10.990	5.66	6.64
nominal	relaxed	FlatRF0.3	2048	10.706	12.446	6.05	7.13
nominal	relaxed	FlatRF3	2048	12.152	10.031	5.81	7.23

Population	Service	Policy	n	L	W	Obs.	Union
nominal	relaxed	KG0.3	2048	11.002	11.389	5.86	6.88
nominal	relaxed	KG1	2048	11.408	10.619	5.76	6.93
nominal	relaxed	KG3	2048	12.031	9.646	5.81	7.08
nominal	relaxed	OldRAVC	2048	11.950	9.600	5.42	6.74
nominal	relaxed	RF_Time	2048	11.125	10.990	5.66	6.64
nominal	relaxed	Random2or4	2048	12.254	10.507	7.42	8.50
nominal	relaxed	Residual	2048	13.590	9.403	6.88	8.45
nominal	standard	CART_Time	2048	11.664	10.705	5.52	6.84
nominal	standard	ContractRF	2048	13.844	9.973	7.18	8.64
nominal	standard	ContractRF0.3	2048	13.844	9.973	7.18	8.64
nominal	standard	ContractRF3	2048	13.844	9.973	7.18	8.64
nominal	standard	Final	2048	11.295	12.109	6.49	7.52
nominal	standard	Fixed16	2048	11.295	12.109	6.49	7.52
nominal	standard	Fixed2	2048	13.833	9.221	7.18	8.64
nominal	standard	Fixed4	2048	11.295	12.109	6.49	7.52
nominal	standard	Fixed8	2048	11.295	12.109	6.49	7.52
nominal	standard	FlatCART	2048	11.664	10.705	5.52	6.84
nominal	standard	FlatCART0.3	2048	11.269	11.495	5.62	6.79
nominal	standard	FlatCART3	2048	12.301	9.721	5.76	7.13
nominal	standard	FlatRF	2048	13.844	9.973	7.18	8.64
nominal	standard	FlatRF0.3	2048	13.844	9.973	7.18	8.64
nominal	standard	FlatRF3	2048	13.844	9.973	7.18	8.64
nominal	standard	KG0.3	2048	11.269	11.495	5.62	6.79
nominal	standard	KG1	2048	11.664	10.705	5.52	6.84
nominal	standard	KG3	2048	12.301	9.721	5.76	7.13
nominal	standard	OldRAVC	2048	12.199	9.666	5.42	6.79
nominal	standard	RF_Time	2048	13.844	9.973	7.18	8.64
nominal	standard	Random2or4	2048	12.379	10.520	7.13	8.25
nominal	standard	Residual	2048	13.817	9.476	6.98	8.50
nominal	tight	CART_Time	2048	17.401	7.522	73.44	73.63
nominal	tight	ContractRF	2048	17.474	7.784	96.44	96.44
nominal	tight	ContractRF0.3	2048	17.474	7.784	96.44	96.44
nominal	tight	ContractRF3	2048	17.474	7.784	96.44	96.44
nominal	tight	Final	2048	17.391	7.511	71.73	71.92
nominal	tight	Fixed16	2048	17.391	7.511	71.73	71.92
nominal	tight	Fixed2	2048	17.391	7.511	71.73	71.92
nominal	tight	Fixed4	2048	17.391	7.511	71.73	71.92
nominal	tight	Fixed8	2048	17.391	7.511	71.73	71.92
nominal	tight	FlatCART	2048	17.401	7.522	73.44	73.63
nominal	tight	FlatCART0.3	2048	17.401	7.522	73.44	73.63
nominal	tight	FlatCART3	2048	17.401	7.522	73.44	73.63
nominal	tight	FlatRF	2048	17.474	7.784	96.44	96.44
nominal	tight	FlatRF0.3	2048	17.474	7.784	96.44	96.44
nominal	tight	FlatRF3	2048	17.474	7.784	96.44	96.44
nominal	tight	KG0.3	2048	17.401	7.522	73.44	73.63
nominal	tight	KG1	2048	17.401	7.522	73.44	73.63
nominal	tight	KG3	2048	17.401	7.522	73.44	73.63
nominal	tight	OldRAVC	2048	17.397	7.519	72.85	73.05
nominal	tight	RF_Time	2048	17.474	7.784	96.44	96.44
nominal	tight	Random2or4	2048	17.391	7.511	71.73	71.92
nominal	tight	Residual	2048	17.391	7.511	71.73	71.92

S7. Complete continuous-route catalogue

Table S6: All 12 methods and nine geometry/condition combinations. Each mean summarizes four matched 600-s inputs. CPU covers the complete online loop; route evidence remains descriptive.

Geometry	Condition	Method	RMSE (cm)	Success (%)	CPU (s)
B767	contention	CART_Time	3.543	54.43	2.038
B767	contention	ContractRF	3.523	44.57	2.345
B767	contention	Final	3.605	45.48	2.566
B767	contention	Fixed2	3.555	46.99	2.322
B767	contention	Fixed4	3.605	45.48	2.607
B767	contention	FlatCART	3.581	54.10	2.505
B767	contention	FlatRF	3.549	44.57	2.576
B767	contention	Full	3.531	55.02	2.019
B767	contention	KG1	3.543	54.43	2.246
B767	contention	KG1_Held	3.545	54.43	2.325
B767	contention	OldRAVC	3.550	54.85	2.299
B767	contention	TaskAwareFlatValidated	3.531	55.02	2.158
B767	stress	CART_Time	8.200	55.85	2.213
B767	stress	ContractRF	8.186	47.74	2.136
B767	stress	Final	8.215	39.97	2.668
B767	stress	Fixed2	8.185	42.22	2.154
B767	stress	Fixed4	8.215	39.97	2.557
B767	stress	FlatCART	8.215	53.85	2.583
B767	stress	FlatRF	8.187	47.41	2.450
B767	stress	Full	8.158	56.19	2.300
B767	stress	KG1	8.200	55.85	2.207
B767	stress	KG1_Held	8.205	55.60	2.199
B767	stress	OldRAVC	8.212	53.18	2.440
B767	stress	TaskAwareFlatValidated	8.158	56.19	2.175
B767	ordinary	CART_Time	3.706	96.49	2.767
B767	ordinary	ContractRF	3.805	97.07	2.728
B767	ordinary	Final	3.719	94.57	3.228
B767	ordinary	Fixed2	3.806	97.07	2.450
B767	ordinary	Fixed4	3.719	94.57	3.176
B767	ordinary	FlatCART	3.706	96.49	2.711
B767	ordinary	FlatRF	3.805	97.07	2.596
B767	ordinary	Full	3.715	97.24	2.514
B767	ordinary	KG1	3.706	96.49	2.809
B767	ordinary	KG1_Held	3.712	96.32	2.562
B767	ordinary	OldRAVC	3.722	97.07	2.500
B767	ordinary	TaskAwareFlatValidated	3.715	97.24	2.504
C172	contention	CART_Time	5.341	80.94	2.162
C172	contention	ContractRF	5.256	53.93	2.090
C172	contention	Final	5.388	51.76	2.734
C172	contention	Fixed2	5.237	53.01	2.121
C172	contention	Fixed4	5.388	51.76	2.659
C172	contention	FlatCART	5.321	81.02	2.374
C172	contention	FlatRF	5.236	53.68	2.288
C172	contention	Full	5.234	81.61	2.028
C172	contention	KG1	5.341	80.94	2.194
C172	contention	KG1_Held	5.342	80.85	2.096
C172	contention	OldRAVC	5.193	81.77	2.173
C172	contention	TaskAwareFlatValidated	5.234	81.61	2.230
C172	stress	CART_Time	14.439	11.45	2.054
C172	stress	ContractRF	14.398	6.27	2.342
C172	stress	Final	14.333	10.03	2.754

Geometry	Condition	Method	RMSE (cm)	Success (%)	CPU (s)
C172	stress	Fixed2	14.488	10.20	2.168
C172	stress	Fixed4	14.333	10.03	2.545
C172	stress	FlatCART	14.494	11.62	2.432
C172	stress	FlatRF	14.490	6.19	2.285
C172	stress	Full	14.414	11.20	2.221
C172	stress	KG1	14.439	11.45	2.198
C172	stress	KG1_Held	14.440	11.45	2.120
C172	stress	OldRAVC	14.403	11.45	2.480
C172	stress	TaskAwareFlatValidated	14.414	11.20	2.189
C172	ordinary	CART_Time	4.832	93.06	2.607
C172	ordinary	ContractRF	4.787	94.15	2.675
C172	ordinary	Final	4.836	91.64	2.941
C172	ordinary	Fixed2	4.787	94.15	2.297
C172	ordinary	Fixed4	4.836	91.64	3.189
C172	ordinary	FlatCART	4.832	93.06	2.378
C172	ordinary	FlatRF	4.787	94.15	2.651
C172	ordinary	Full	4.805	93.81	2.518
C172	ordinary	KG1	4.832	93.06	2.589
C172	ordinary	KG1_Held	4.832	93.06	2.562
C172	ordinary	OldRAVC	4.837	93.65	2.267
C172	ordinary	TaskAwareFlatValidated	4.805	93.81	2.491
Unfitted	contention	CART_Time	2.993	43.65	2.027
Unfitted	contention	ContractRF	2.993	40.38	2.319
Unfitted	contention	Final	3.071	39.21	2.694
Unfitted	contention	Fixed2	3.064	39.21	2.108
Unfitted	contention	Fixed4	3.071	39.21	2.353
Unfitted	contention	FlatCART	3.054	43.73	2.382
Unfitted	contention	FlatRF	3.057	40.22	2.322
Unfitted	contention	Full	2.994	43.65	1.975
Unfitted	contention	KG1	2.993	43.65	2.078
Unfitted	contention	KG1_Held	2.996	43.65	1.934
Unfitted	contention	OldRAVC	3.048	43.81	2.383
Unfitted	contention	TaskAwareFlatValidated	2.994	43.65	2.090
Unfitted	stress	CART_Time	9.649	46.49	2.207
Unfitted	stress	ContractRF	9.633	42.56	1.907
Unfitted	stress	Final	9.283	37.04	2.531
Unfitted	stress	Fixed2	9.665	38.21	2.017
Unfitted	stress	Fixed4	9.283	37.04	2.519
Unfitted	stress	FlatCART	9.672	45.07	2.500
Unfitted	stress	FlatRF	9.661	42.14	2.451
Unfitted	stress	Full	9.663	45.74	2.211
Unfitted	stress	KG1	9.649	46.49	2.147
Unfitted	stress	KG1_Held	9.654	46.49	2.042
Unfitted	stress	OldRAVC	9.269	44.82	2.464
Unfitted	stress	TaskAwareFlatValidated	9.663	45.74	2.401
Unfitted	ordinary	CART_Time	3.018	99.25	2.507
Unfitted	ordinary	ContractRF	2.997	99.41	2.673
Unfitted	ordinary	Final	3.022	98.91	2.674
Unfitted	ordinary	Fixed2	2.997	99.41	2.413
Unfitted	ordinary	Fixed4	3.022	98.91	2.688
Unfitted	ordinary	FlatCART	3.018	99.25	2.635
Unfitted	ordinary	FlatRF	2.997	99.41	2.521
Unfitted	ordinary	Full	3.006	99.25	2.395
Unfitted	ordinary	KG1	3.018	99.25	2.618
Unfitted	ordinary	KG1_Held	3.027	99.16	2.577

Geometry	Condition	Method	RMSE (cm)	Success (%)	CPU (s)
Unfitted	ordinary	OldRAVC	3.006	99.25	2.290
Unfitted	ordinary	TaskAwareFlatValidated	3.006	99.25	2.235

S8. Image-task numerical confirmation and stress

Table S7: All eight numerical policies by geometry for the standard service. Total work includes the virtual camera. The complete 192-row all-service table is supplied as CSV; geometry summaries carry no conditional risk certificate.

Study	Geometry	Policy	n	RMSE (cm)	Work (ms)	Union (%)
A	B767	ContractRF	515	4.872	72.816	9.71
A	B767	Final	515	4.147	74.974	7.77
A	B767	Fixed2	515	4.872	72.074	9.71
A	B767	Fixed4	515	4.147	74.974	7.77
A	B767	FlatCART	515	4.327	73.629	6.60
A	B767	FlatRF	515	4.872	72.816	9.71
A	B767	KG1	515	4.327	73.629	6.60
A	B767	OldRAVC	515	4.624	72.595	7.77
A	C172	ContractRF	509	4.609	72.087	6.48
A	C172	Final	509	4.378	74.237	6.68
A	C172	Fixed2	509	4.609	71.330	6.48
A	C172	Fixed4	509	4.378	74.237	6.68
A	C172	FlatCART	509	4.503	72.568	6.09
A	C172	FlatRF	509	4.609	72.087	6.48
A	C172	KG1	509	4.503	72.568	6.09
A	C172	OldRAVC	509	4.571	71.654	6.09
A	Unfitted	ContractRF	1024	4.864	72.454	10.45
A	Unfitted	Final	1024	4.499	74.263	9.47
A	Unfitted	Fixed2	1024	4.864	71.719	10.45
A	Unfitted	Fixed4	1024	4.499	74.263	9.47
A	Unfitted	FlatCART	1024	4.640	73.235	9.57
A	Unfitted	FlatRF	1024	4.864	72.454	10.45
A	Unfitted	KG1	1024	4.640	73.235	9.57
A	Unfitted	OldRAVC	1024	4.728	72.249	9.28
B	B767	ContractRF	513	5.164	135.336	12.48
B	B767	Final	513	4.333	137.553	9.36
B	B767	Fixed2	513	5.164	134.596	12.48
B	B767	Fixed4	513	4.333	137.553	9.36
B	B767	FlatCART	513	4.495	136.455	8.38
B	B767	FlatRF	513	5.164	135.336	12.48
B	B767	KG1	513	4.495	136.455	8.38
B	B767	OldRAVC	513	4.705	135.184	8.58
B	C172	ContractRF	511	4.346	134.588	3.33
B	C172	Final	511	4.154	136.753	4.70
B	C172	Fixed2	511	4.346	133.831	3.33
B	C172	Fixed4	511	4.154	136.753	4.70
B	C172	FlatCART	511	4.326	135.248	3.72
B	C172	FlatRF	511	4.346	134.588	3.33
B	C172	KG1	511	4.326	135.248	3.72
B	C172	OldRAVC	511	4.315	134.111	2.94
B	Unfitted	ContractRF	1024	4.829	134.947	10.55

Study	Geometry	Policy	n	RMSE (cm)	Work (ms)	Union (%)
B	Unfitted	Final	1024	4.427	136.759	8.59
B	Unfitted	Fixed2	1024	4.829	134.212	10.55
B	Unfitted	Fixed4	1024	4.427	136.759	8.59
B	Unfitted	FlatCART	1024	4.552	135.694	8.79
B	Unfitted	FlatRF	1024	4.829	134.947	10.55
B	Unfitted	KG1	1024	4.552	135.694	8.79
B	Unfitted	OldRAVC	1024	4.663	134.805	9.38

Table S8: All standard-service unsupported image stress strata and policies. The full 512-row all-service table is retained as CSV. These observations carry no transferred population-risk statement.

Study	Stress	Geometry	Policy	n	RMSE (cm)	Union (%)
A	degeneracy	B767	ContractRF	28	5.92	14.3
A	degeneracy	B767	Final	28	5.23	17.9
A	degeneracy	B767	Fixed2	28	5.92	14.3
A	degeneracy	B767	Fixed4	28	5.23	17.9
A	degeneracy	B767	FlatCART	28	5.22	17.9
A	degeneracy	B767	FlatRF	28	5.92	14.3
A	degeneracy	B767	KG1	28	5.22	17.9
A	degeneracy	B767	OldRAVC	28	5.80	14.3
A	degeneracy	C172	ContractRF	36	3.70	2.8
A	degeneracy	C172	Final	36	4.05	5.6
A	degeneracy	C172	Fixed2	36	3.70	2.8
A	degeneracy	C172	Fixed4	36	4.05	5.6
A	degeneracy	C172	FlatCART	36	3.78	2.8
A	degeneracy	C172	FlatRF	36	3.70	2.8
A	degeneracy	C172	KG1	36	3.78	2.8
A	degeneracy	C172	OldRAVC	36	3.72	2.8
A	map_shift	B767	ContractRF	29	7.01	20.7
A	map_shift	B767	Final	29	6.44	27.6
A	map_shift	B767	Fixed2	29	7.01	20.7
A	map_shift	B767	Fixed4	29	6.44	27.6
A	map_shift	B767	FlatCART	29	6.50	20.7
A	map_shift	B767	FlatRF	29	7.01	20.7
A	map_shift	B767	KG1	29	6.50	20.7
A	map_shift	B767	OldRAVC	29	6.80	20.7
A	map_shift	C172	ContractRF	35	11.05	82.9
A	map_shift	C172	Final	35	10.77	85.7
A	map_shift	C172	Fixed2	35	11.05	82.9
A	map_shift	C172	Fixed4	35	10.77	85.7
A	map_shift	C172	FlatCART	35	10.75	88.6
A	map_shift	C172	FlatRF	35	11.05	82.9
A	map_shift	C172	KG1	35	10.75	88.6
A	map_shift	C172	OldRAVC	35	11.11	85.7
A	occlusion	B767	ContractRF	35	5.63	28.6
A	occlusion	B767	Final	35	4.84	20.0
A	occlusion	B767	Fixed2	35	5.63	28.6
A	occlusion	B767	Fixed4	35	4.84	20.0
A	occlusion	B767	FlatCART	35	4.88	22.9
A	occlusion	B767	FlatRF	35	5.63	28.6
A	occlusion	B767	KG1	35	4.88	22.9

Study	Stress	Geometry	Policy	n	RMSE (cm)	Union (%)
A	occlusion	B767	OldRAVC	35	5.04	22.9
A	occlusion	C172	ContractRF	29	5.94	31.0
A	occlusion	C172	Final	29	5.68	31.0
A	occlusion	C172	Fixed2	29	5.94	31.0
A	occlusion	C172	Fixed4	29	5.68	31.0
A	occlusion	C172	FlatCART	29	5.92	31.0
A	occlusion	C172	FlatRF	29	5.94	31.0
A	occlusion	C172	KG1	29	5.92	31.0
A	occlusion	C172	OldRAVC	29	5.94	31.0
A	prior_shift	B767	ContractRF	30	7.16	26.7
A	prior_shift	B767	Final	30	4.33	6.7
A	prior_shift	B767	Fixed2	30	7.16	26.7
A	prior_shift	B767	Fixed4	30	4.33	6.7
A	prior_shift	B767	FlatCART	30	4.19	6.7
A	prior_shift	B767	FlatRF	30	7.16	26.7
A	prior_shift	B767	KG1	30	4.19	6.7
A	prior_shift	B767	OldRAVC	30	5.35	13.3
A	prior_shift	C172	ContractRF	34	7.10	29.4
A	prior_shift	C172	Final	34	5.55	14.7
A	prior_shift	C172	Fixed2	34	7.10	29.4
A	prior_shift	C172	Fixed4	34	5.55	14.7
A	prior_shift	C172	FlatCART	34	5.61	14.7
A	prior_shift	C172	FlatRF	34	7.10	29.4
A	prior_shift	C172	KG1	34	5.61	14.7
A	prior_shift	C172	OldRAVC	34	6.27	20.6
B	degeneracy	B767	ContractRF	31	9.20	32.3
B	degeneracy	B767	Final	31	7.36	22.6
B	degeneracy	B767	Fixed2	31	9.20	32.3
B	degeneracy	B767	Fixed4	31	7.36	22.6
B	degeneracy	B767	FlatCART	31	7.70	25.8
B	degeneracy	B767	FlatRF	31	9.20	32.3
B	degeneracy	B767	KG1	31	7.70	25.8
B	degeneracy	B767	OldRAVC	31	8.36	29.0
B	degeneracy	C172	ContractRF	33	4.38	6.1
B	degeneracy	C172	Final	33	4.11	9.1
B	degeneracy	C172	Fixed2	33	4.38	6.1
B	degeneracy	C172	Fixed4	33	4.11	9.1
B	degeneracy	C172	FlatCART	33	4.22	6.1
B	degeneracy	C172	FlatRF	33	4.38	6.1
B	degeneracy	C172	KG1	33	4.22	6.1
B	degeneracy	C172	OldRAVC	33	4.38	6.1
B	map_shift	B767	ContractRF	26	8.44	34.6
B	map_shift	B767	Final	26	7.87	30.8
B	map_shift	B767	Fixed2	26	8.44	34.6
B	map_shift	B767	Fixed4	26	7.87	30.8
B	map_shift	B767	FlatCART	26	7.83	30.8
B	map_shift	B767	FlatRF	26	8.44	34.6
B	map_shift	B767	KG1	26	7.83	30.8
B	map_shift	B767	OldRAVC	26	7.88	30.8
B	map_shift	C172	ContractRF	38	10.39	44.7
B	map_shift	C172	Final	38	10.06	52.6
B	map_shift	C172	Fixed2	38	10.39	44.7
B	map_shift	C172	Fixed4	38	10.06	52.6
B	map_shift	C172	FlatCART	38	10.04	52.6
B	map_shift	C172	FlatRF	38	10.39	44.7

Study	Stress	Geometry	Policy	n	RMSE (cm)	Union (%)
B	map_shift	C172	KG1	38	10.04	52.6
B	map_shift	C172	OldRAVC	38	10.07	44.7
B	occlusion	B767	ContractRF	34	6.48	35.3
B	occlusion	B767	Final	34	5.69	35.3
B	occlusion	B767	Fixed2	34	6.48	35.3
B	occlusion	B767	Fixed4	34	5.69	35.3
B	occlusion	B767	FlatCART	34	6.09	35.3
B	occlusion	B767	FlatRF	34	6.48	35.3
B	occlusion	B767	KG1	34	6.09	35.3
B	occlusion	B767	OldRAVC	34	5.96	35.3
B	occlusion	C172	ContractRF	30	4.72	16.7
B	occlusion	C172	Final	30	4.63	20.0
B	occlusion	C172	Fixed2	30	4.72	16.7
B	occlusion	C172	Fixed4	30	4.63	20.0
B	occlusion	C172	FlatCART	30	4.90	20.0
B	occlusion	C172	FlatRF	30	4.72	16.7
B	occlusion	C172	KG1	30	4.90	20.0
B	occlusion	C172	OldRAVC	30	4.72	16.7
B	prior_shift	B767	ContractRF	29	5.70	10.3
B	prior_shift	B767	Final	29	5.11	13.8
B	prior_shift	B767	Fixed2	29	5.70	10.3
B	prior_shift	B767	Fixed4	29	5.11	13.8
B	prior_shift	B767	FlatCART	29	5.12	10.3
B	prior_shift	B767	FlatRF	29	5.70	10.3
B	prior_shift	B767	KG1	29	5.12	10.3
B	prior_shift	B767	OldRAVC	29	5.23	6.9
B	prior_shift	C172	ContractRF	35	6.47	17.1
B	prior_shift	C172	Final	35	4.85	8.6
B	prior_shift	C172	Fixed2	35	6.47	17.1
B	prior_shift	C172	Fixed4	35	4.85	8.6
B	prior_shift	C172	FlatCART	35	4.95	8.6
B	prior_shift	C172	FlatRF	35	6.47	17.1
B	prior_shift	C172	KG1	35	4.95	8.6
B	prior_shift	C172	OldRAVC	35	5.35	5.7

S9. Additional diagnostic figures retained from the executed studies

These figures preserve the all-service, tail and geometry detail moved out of the main narrative. Their observations, sampling units and frozen statistical allocations are unchanged.

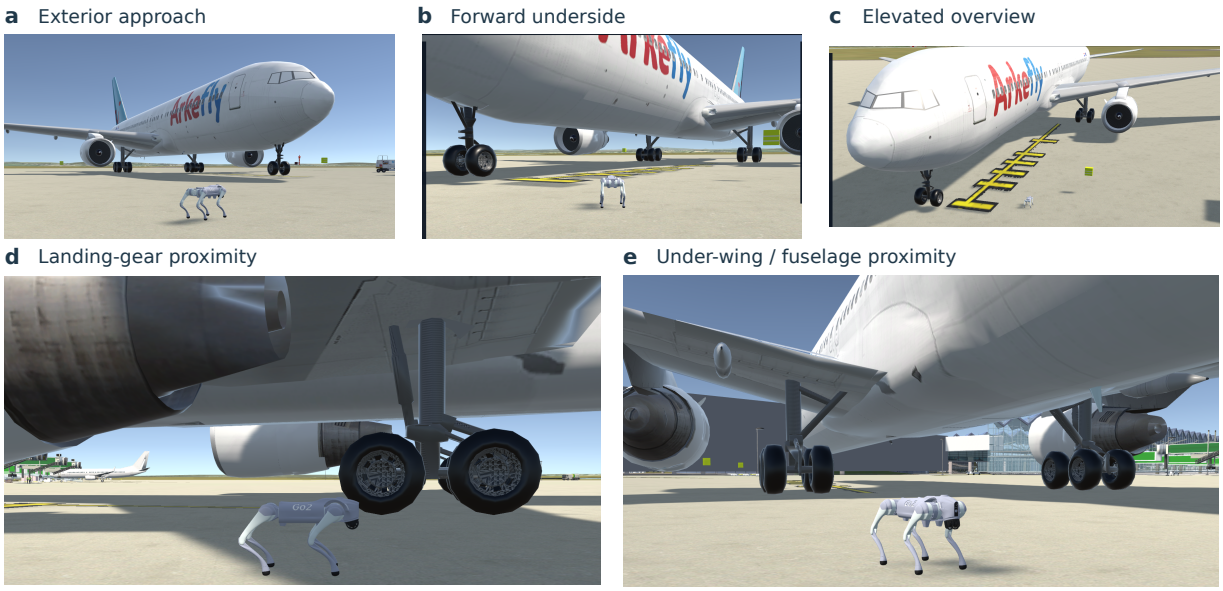


Figure S1: The five originally supplied simulation views, preserved as qualitative geometry and platform context. They are distinct from the independently generated episode and inspection-task samples.

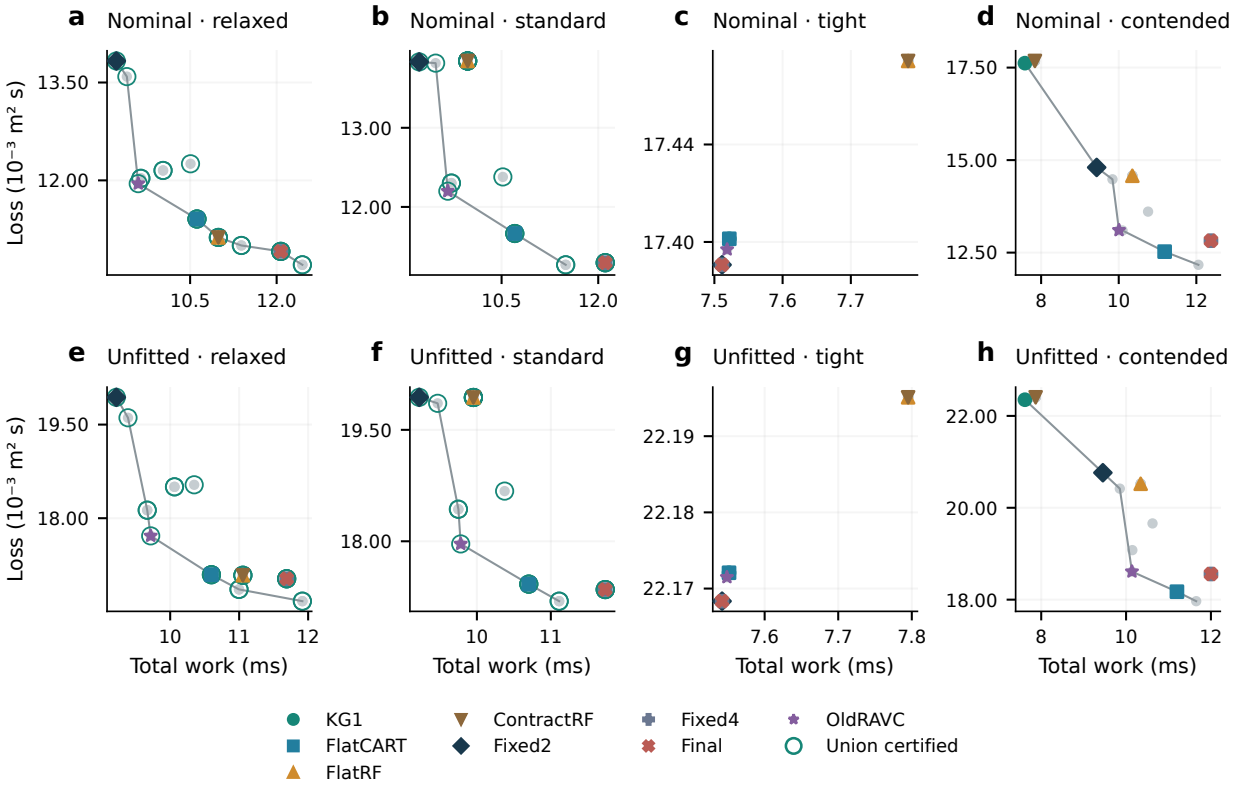


Figure S2: All 22 localization policy means across both confirmation populations and four services. Rings mark direct-union eligibility; Pareto lines are descriptive.

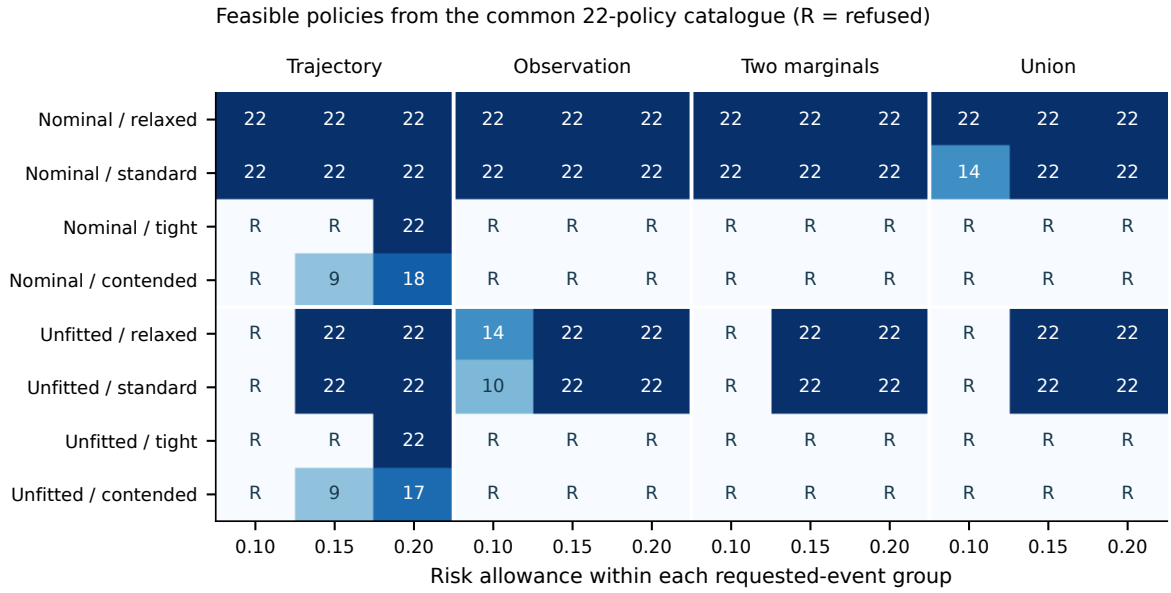


Figure S3: All 96 frozen localization requests across task and allowance variants. Empty catalogues are retained as refusals.

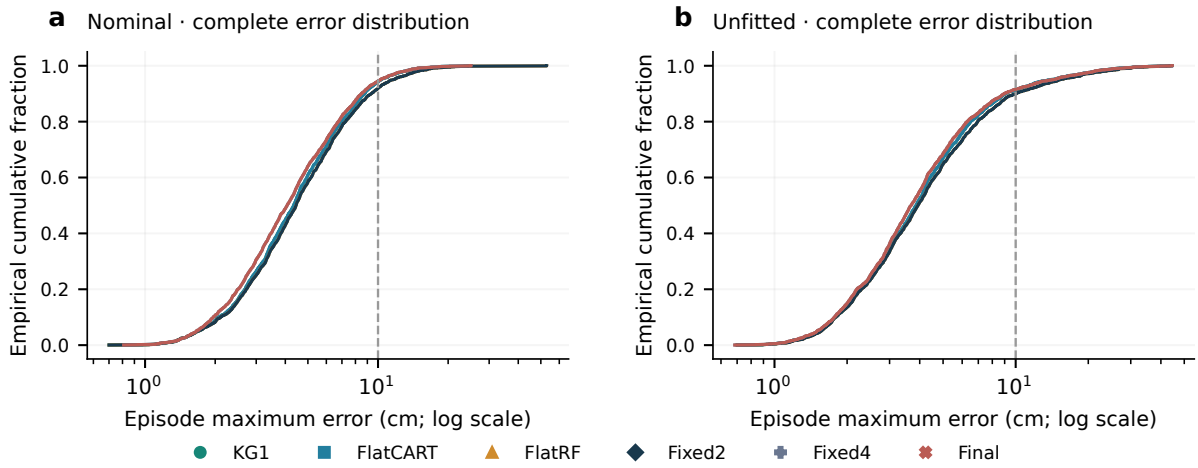


Figure S4: Empirical distributions of all confirmation episode maxima. The reference is 10 cm.

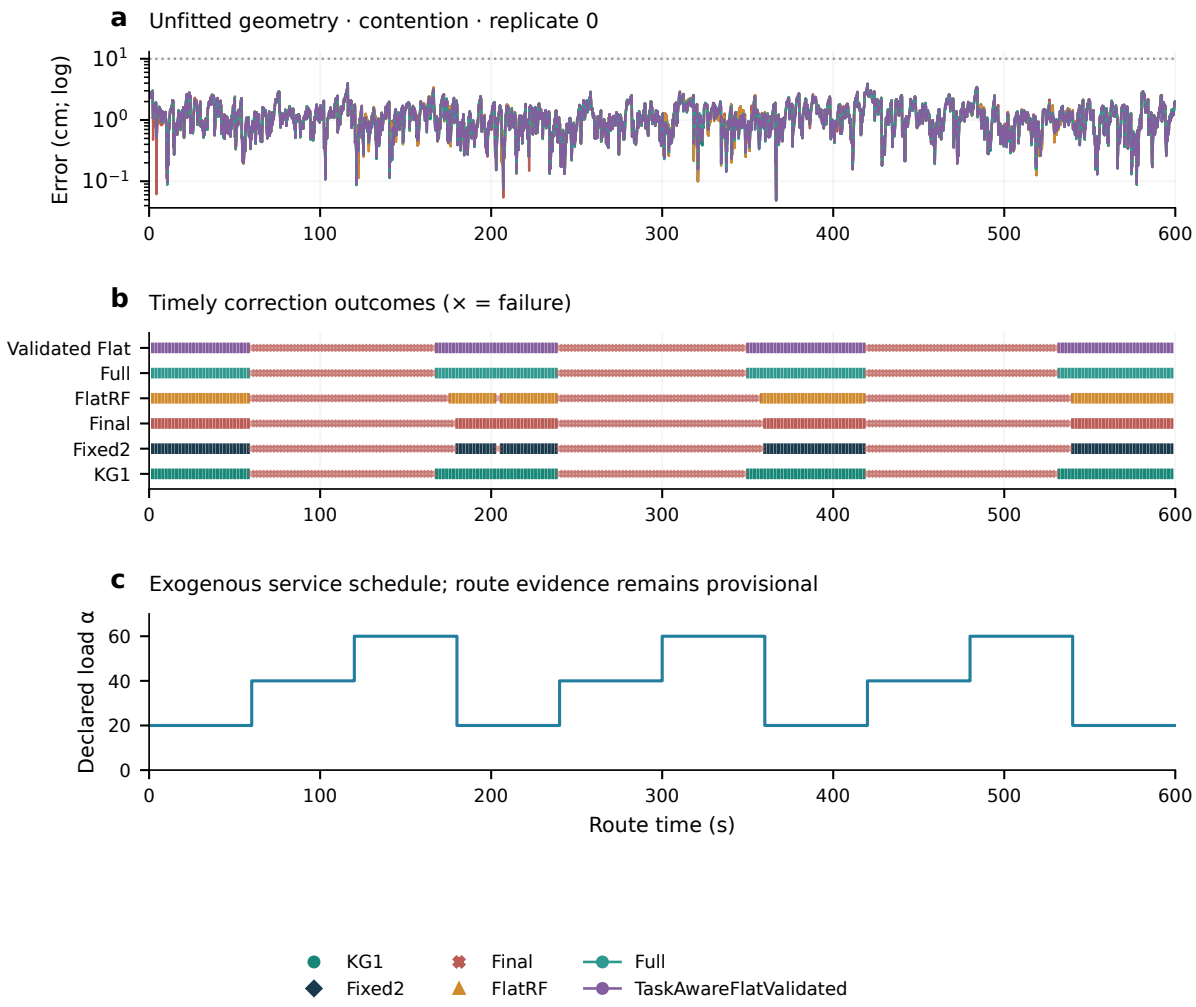


Figure S5: Predeclared unfitted-airframe/contention/replicate-zero route: all 6,000 position-error ticks, correction opportunities and exogenous processor load.

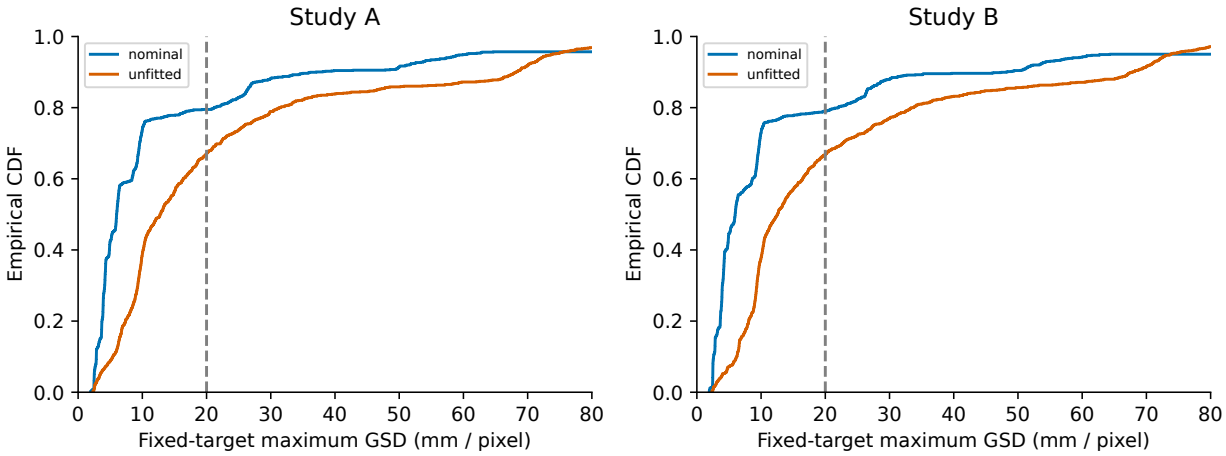


Figure S6: Fixed-target GSD empirical distributions on independent A/B confirmation samples. Planned targets outside the acquired ROI remain in the geometric score.

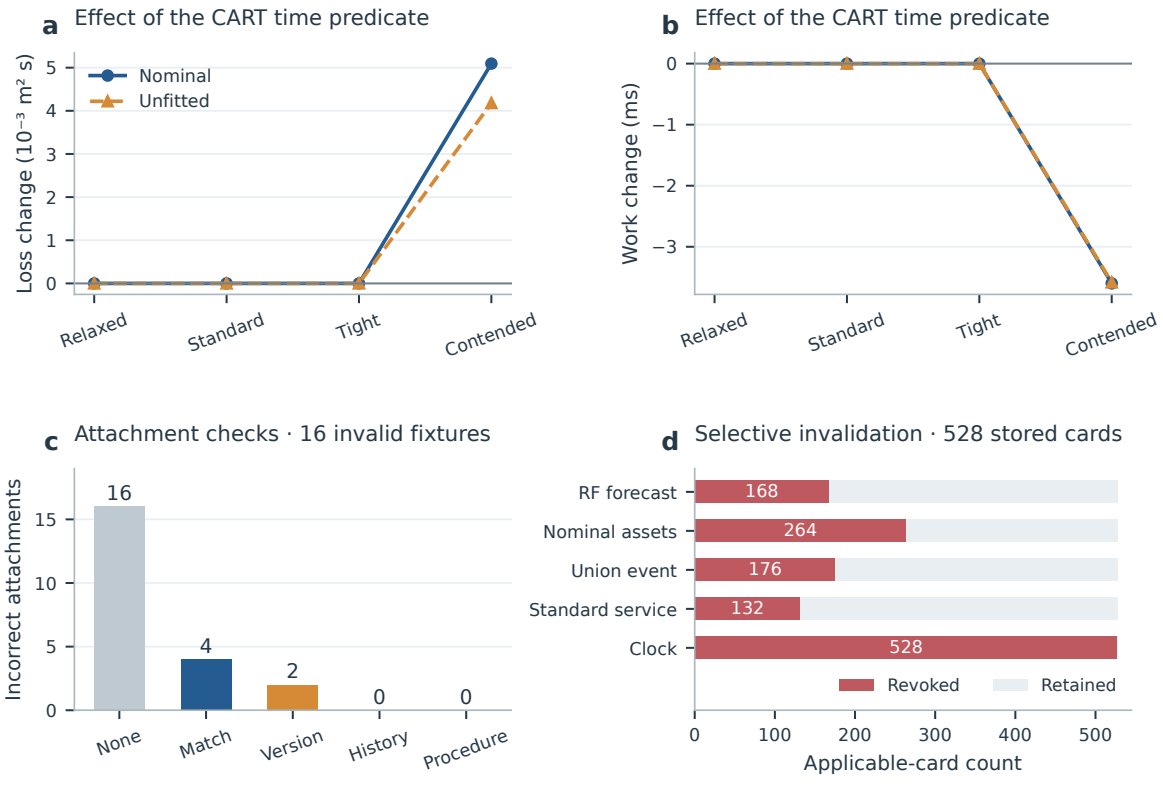
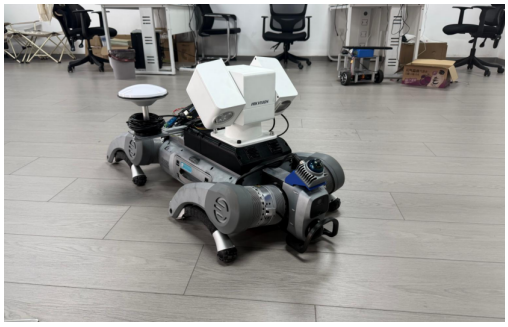


Figure S7: Localization mechanism diagnostics. Paired time-predicate effects across four services and two populations; incorrect attachments across 16 fixed invalid fixtures; and selective revocation among the original 528 localization cards. These counts retain their distinct original denominators.

a Laboratory platform



b Sensor integration



c Aircraft field setting



d Under-fuselage setting



Figure S8: The four original platform photographs, retained as project engineering context.

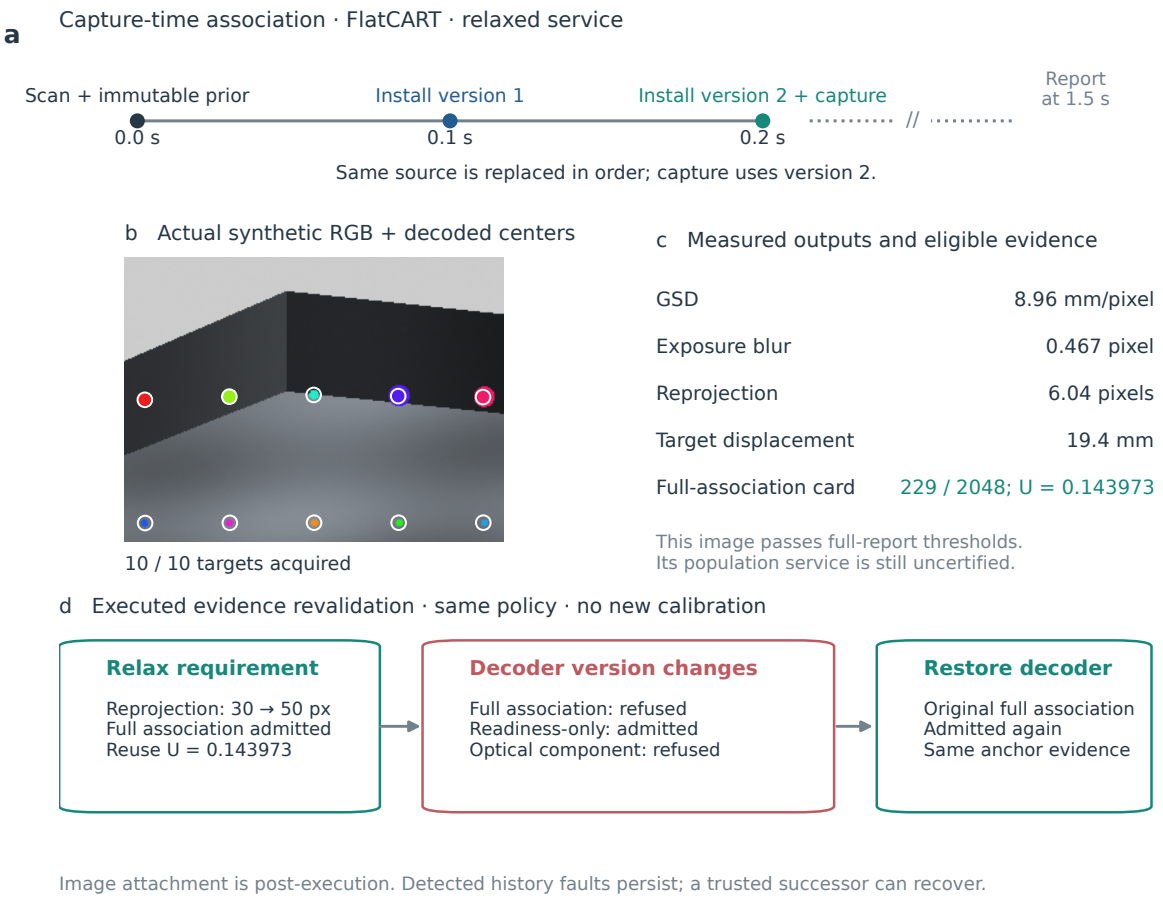


Figure S9: Original pixel-level diagnostic image, execution trace and evidence revalidation case. The raster is the actual archived synthetic output.

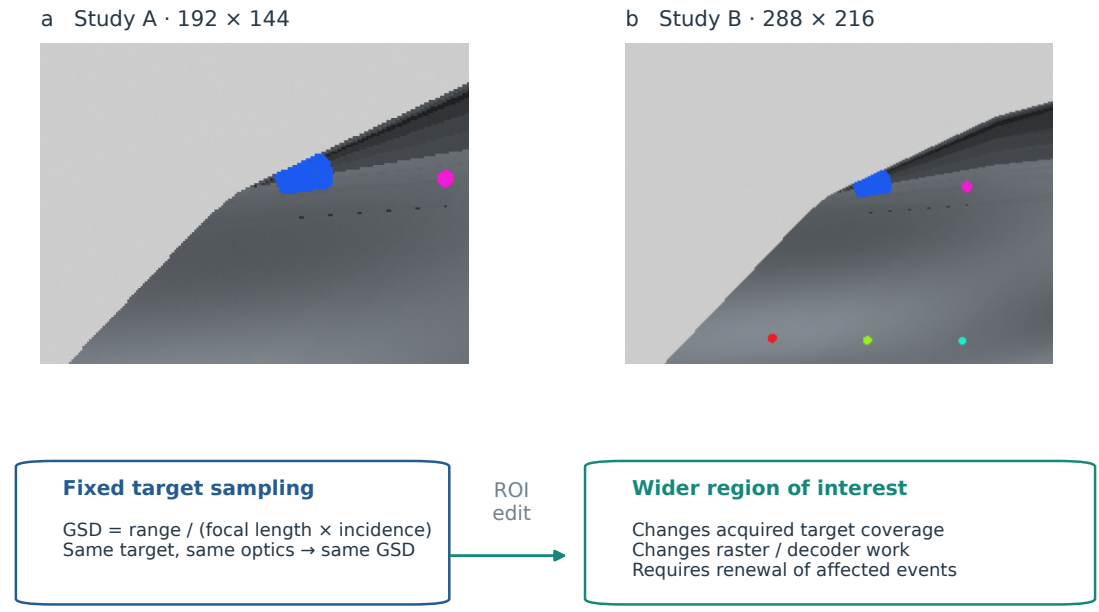
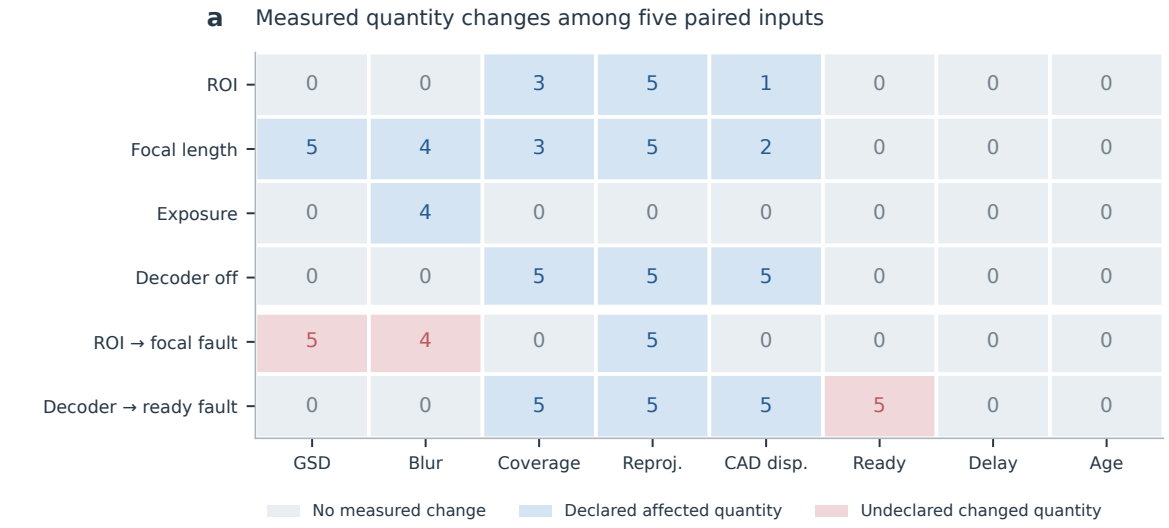


Figure S10: Original paired-ROI raster control with fixed CAD landmarks and optical parameters.



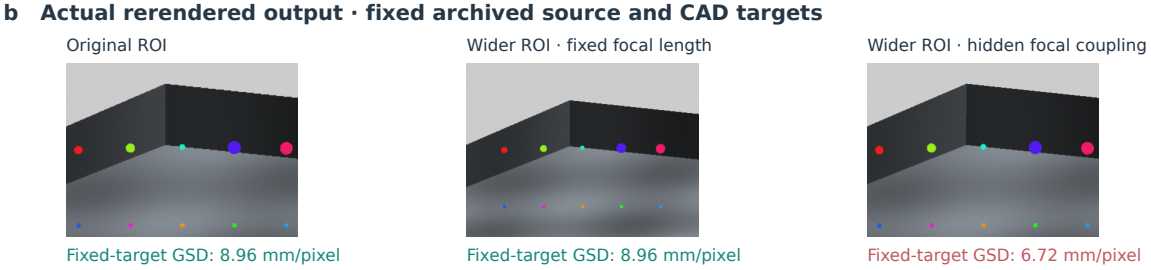


Figure S11: Original five-template producer re-execution, preceding the controlled camera campaign.

S10. Typed metamodel, compilation and executed producer auditing

The reference extension implements typed request compilation and a relation-domain checker. A request contains population, service, allowance and a conjunction of intervals. Each interval declares a metric and engineering unit, with optional lower and upper bounds. Unit conversion, interval intersection and canonical sorting produce the event identity; contradictory bounds and incompatible dimensions are rejected. All 24 declared task/variant expressions reproduce their original event identities and provider projections. Four invalid quantity specifications, a wrong relation type and an invalid source-revision order are rejected.

Table S9: Metamodel relation signatures implemented by the checker.

Relation	Domain	Range
requires	Request	Event
commits	Plan	Request
selects	Plan	Policy
bounds	Card	Event
supports	Card	Plan
derivedFrom	Card	Producer
produces	Producer	Quantity
dependsOn	Quantity	Provider
uses	Receipt	SourceVersion
records	Receipt	Plan
revises	SourceVersion	SourceVersion

For example, full association compiles coverage ≥ 0.50, reprojection ≤ 30 px, CAD-target displacement ≤ 100 mm and readiness ≥ 1. The compiler converts 100 mm to 0.10 m and gathers the declared dependencies of those quantities. The dependency manifest is an explicit implemented-data-flow specification; it is not inferred by a static-analysis engine. Unit/schema checks validate its use, while paired numerical re-execution audits its consequences.

The five sources are the previously sealed reproduction inputs: A/nominal 13,200,000 and 13,201,023; A/unfitted 13,300,000; B/nominal 14,200,000; and B/unfitted 14,300,000. For each, original CAD addresses, geometry perturbation, motion, capture time and installed FlatCART/relaxed state are retained. The renderer, pixel decoder and task scorer run on these fixed inputs. Baseline RGB arrays agree byte-for-byte with their archives and task quantities agree within 10^{-10}. Four interventions widen ROI by 96 by 72 pixels, set focal length to 2,400 px, set exposure to 9 ms, or disable the decoder. Two fault fixtures additionally make focal length depend on ROI width, or make capture readiness depend on the decoder's finite outputs, without declaring that coupling. The fixed virtual reporting clock is retained to isolate producer-function changes.

Measured quantity changes are computed from output values, independently of the dependency checker. A declared affected quantity may show no change on a finite input; this is not evidence of global invariance. An undeclared measured change falsifies the transfer premise for that tested producer/input. The replay audit flags all five pairs in each injected coupling class. These fixtures test the specified faults and do not estimate arbitrary hidden-coupling detection rates.

S10.1. Reuse-first routing and calibration-scope sensitivity

The routed evidence analysis uses all 320 unique primary contexts. It fixes the originally selected direct-renewal policy, selects a compatible covering anchor when available and otherwise retains an exact-event card. All applicable invariant evidence pools are available to both routes. Selected proofs are deduplicated by event, producer, population and calibration-source manifest; repeated service labels do not add distinct proofs. The catalogue counts measure selected proof-construction requirements, excluding acquisition, producer execution, policy-catalogue search, native timing and human effort.

Canonical calibration scopes use the same four signatures and do not merge tests using observed numerical coincidence. The anchor family contains 780 canonical tests; the complete stored family contains 1,692. Each

Table S10: Actual producer re-execution. Counts refer to five paired archived inputs per intervention.

Intervention	Pairs	Undeclared changes	Flagged pairs
roi	5	0	0
focal	5	0	0
exposure	5	0	0
decoder_disabled	5	0	0
hidden_roi_focal	5	9	5
hidden_decoder_readiness	5	5	5

Table S11: Selected proof preparation on identical supported contexts and fixed numerical policies.

Study	Route	Contexts	Proof cards	Record evaluations
A	Exact event	30	18	36864
A	Reuse + fallback	30	15	30720
B	Exact event	29	17	36864
B	Reuse + fallback	29	15	32768
combined	Exact event	59	35	73728
combined	Reuse + fallback	59	30	63488

sensitivity design separately allocates total confidence error 0.05. A combined routed catalogue requires one allocation over its tested union. The original primary cards continue to use 0.05/4,608. The canonical designs tighten the example bound without changing support at allowance 0.15.

Table S12: Archived-count scope sensitivity for the full-association example, with 229 failures among 2,048 calibration inputs.

Scope design	Method	Family size	Upper bound
Primary reserved	Anchor	4608	0.143973
Primary reserved	Direct	4608	0.143973
Canonical scope design	Anchor	780	0.140651
Canonical scope design	Direct	1692	0.142133

S10.2. Registration point-count probes

The unchanged source solver is run on the existing frontend cloud at requested caps 96, 192 and 384. Actual sample count clips to available points. Each run uses the original immutable prior; no forecast model is retrained. Native Linux CPU is the median of three repeated executions, including solver construction. The error column describes the final candidate before acceptance; rejected candidates remain visible. These finite probes use no primary budget wrapper or larger-point risk certificate.

Table S13: Paired point-count diagnostic on the five archived inputs. Timing repeats do not increase the source count.

Source	Cap	Used	Iterations	Accepted	Error (mm)	CPU (ms)
13200000	96	96	5	no	82.68	14.68
13200000	192	189	5	yes	88.23	16.01
13200000	384	189	5	yes	88.23	15.47
13201023	96	96	4	yes	81.79	8.71
13201023	192	192	6	yes	39.32	14.39
13201023	384	384	5	yes	73.40	14.40
13300000	96	96	3	yes	27.74	8.80
13300000	192	192	5	yes	11.29	11.83
13300000	384	205	5	yes	13.56	14.38
14200000	96	96	5	no	147.08	12.82
14200000	192	192	7	no	83.13	16.82
14200000	384	384	8	yes	87.18	17.85
14300000	96	96	3	yes	19.38	6.09
14300000	192	192	3	yes	16.04	7.69
14300000	384	384	3	yes	28.17	9.76

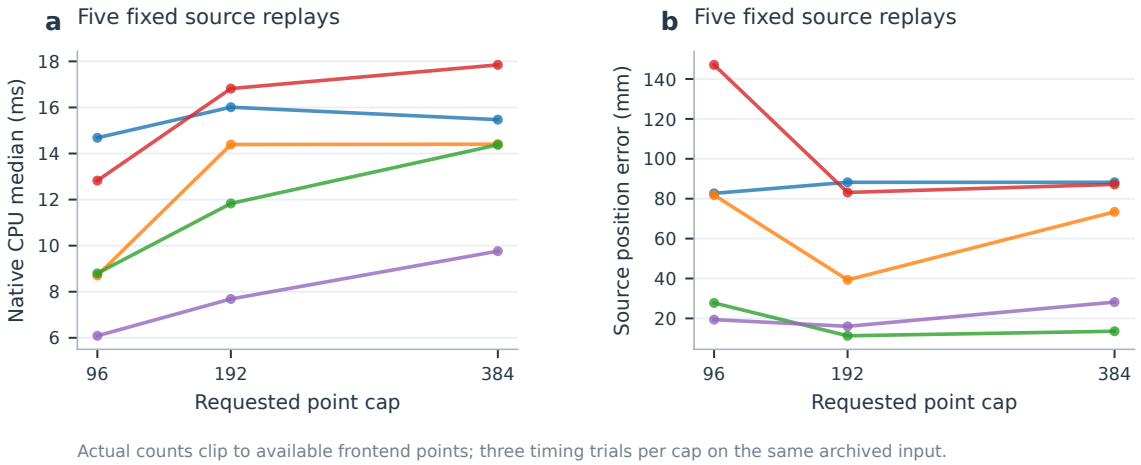


Figure S12: Executed point-count probes. Every trace is one fixed archived source. Candidate error is nonmonotonic in requested point count; timing medians describe this Linux diagnostic.

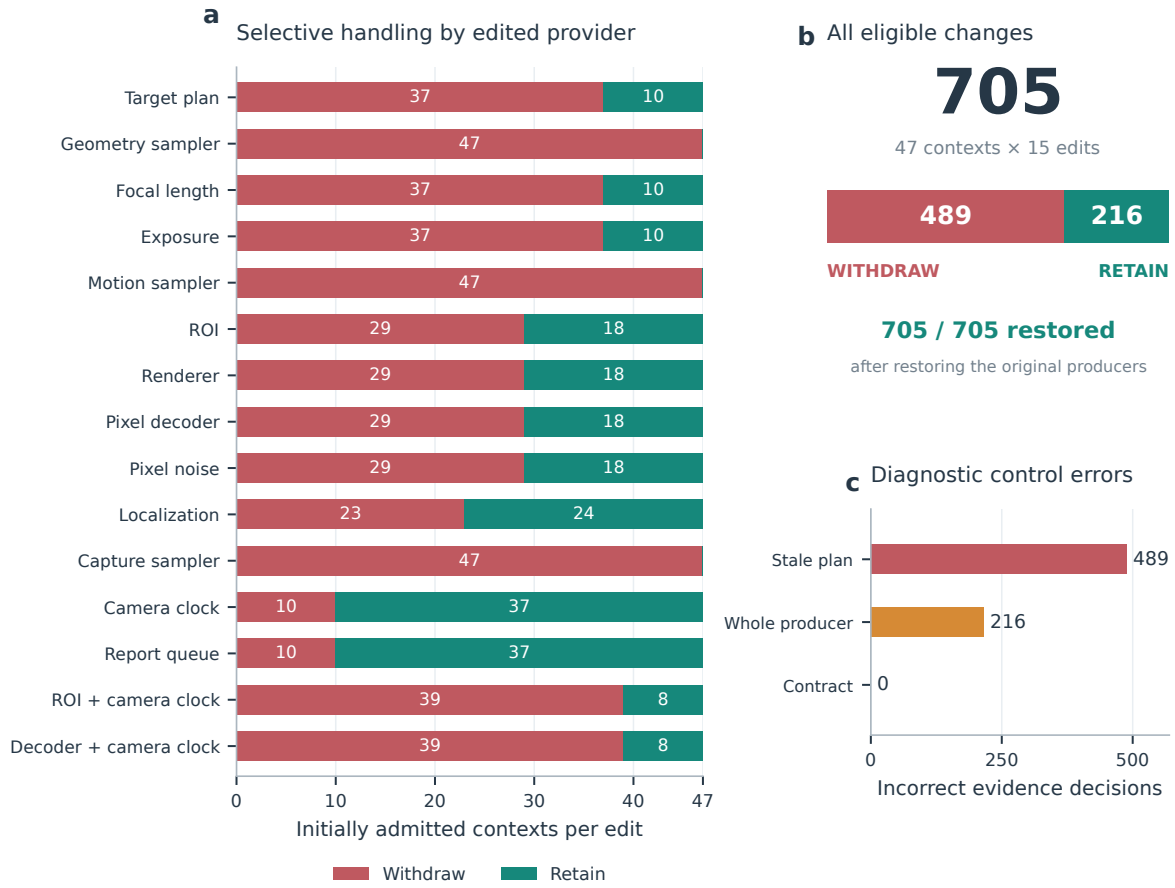


Figure S13: Original declared-model change/recovery conformance matrix, retained as implementation verification. The oracle shares the declaration specification; these counts do not test the correctness of an undeclared physical dependency.

S11. Controlled camera-change campaign

Five archived CAD/capture templates supply nominal A sources 13200000 and 13201023, unfitted A source 13300000, nominal B source 14200000 and unfitted B source 14300000. Each supplies 64 independently seeded camera perturbations. These are simulated configurations conditional on the templates; they add no population calibration cards. Camera translation and rotation have standard deviations 0.04 m and 0.001 rad. Focal length, exposure and brightness are uniform over 1600–2000 pixels, 4–8 ms and 0.70–1.15. Baseline and expanded ROIs are 384 by 288 and 512 by 384. Existing CAD addresses and installed localization estimates are retained.

The paired producer executes geometry, BVH raycasting, Gaussian exposure filtering, pixel decoding and task scoring. Focal-length updates multiply the original value by 1.20; exposure updates multiply it by 1.50; decoder updates suppress decoded output. Each pair uses the same noise seed. ROI–focal faults multiply focal length by 1.15 when the original focal length is at least 1800 pixels. Decoder–readiness faults make readiness depend on decoded output when exposure is at least 6 ms. Exposure–focal faults multiply focal length by 1.03 when edited exposure exceeds 10 ms. ROI–clock faults scale the camera operation clock with ROI area. Each faulty update advertises only its initiating provider change.

Table S14: Executed updates, conditional activation and regression vetoes.

Update	Pairs	Active faults	Vetoes	Undeclared changes
ROI expansion	320	0	0	0
Focal length	320	0	0	0
Exposure	320	0	0	0
Decoder off	320	0	0	0
ROI–focal coupling	320	163	163	288
Decoder–readiness coupling	320	160	160	160
Exposure–focal coupling	320	104	104	104
ROI–clock coupling	320	320	320	640

The configurations, all 20,480 paired quantity comparisons, change-case decisions and output summaries are supplied as CSV. The frozen script and input hashes permit standalone regeneration. Numerical verification compares recomputed values at tolerance 1e-10; CPU timing is not used as the detection criterion. Original primary risk cards and policy plans remain unchanged.